In [8]:
```
%pip install --upgrade --quiet torch pandas numpy matplotlib scikit-learn seaborn
```

```
Note: you may need to restart the kernel to use updated packages.
[notice] A new release of pip is available: 24.0 -> 26.0.1
[notice] To update, run: python.exe -m pip install --upgrade pip
```

Report Overview

This notebook contains the changes implemented at the end of the V1 Report. Those change were

1. Adding Numeric Scaling functionality for any numeric feature
2. Implementing Early Stopping
3. Creating a model which can handle dropout layers
4. Calculating class weights and passing those as input to the loss function
5. Adding the ability to handle custom embedding dimension sizes for categorical features.

Comments are left throughout the notebook where code differs from V1. At the end, I do a breakdown of what the results were for the different changes implemented, and some plans for the next model iteration.

In [9]:
```
import torch
import torch.nn as nn
import torch.optim as optim
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
from sklearn.model_selection import train_test_split
from sklearn.preprocessing import StandardScaler
from torch.utils.data import Dataset, DataLoader
from pathlib import Path
from datetime import datetime
import seaborn as sns
```

Data Preparation

In [10]:
```
data = pd.read_csv('rnn_data.csv')
```

In [11]:
```
data.head()
```

Out[11]:

	pitch_type	game_date	batter	pitcher	description	zone	stand	p_throws	balls	s
0	FF	2025-03-18	660271	684007	called_strike	12.0	L	L	0	
1	SL	2025-03-18	660271	684007	ball	13.0	L	L	0	
2	FF	2025-03-18	660271	684007	hit_into_play	5.0	L	L	1	
3	FS	2025-03-18	669242	684007	ball	13.0	R	L	0	
4	FS	2025-03-18	669242	684007	swinging_strike	14.0	R	L	1	

5 rows × 40 columns

In [12]:
```
data["is_real_pitch"] =  data["pitch_type"].notna() & (data["pitch_type"] != "ABS")

data["target_is_real_pitch"] = data.groupby("pa_id")["is_real_pitch"].shift(-1)

data["y_next_pitch_type"] = data.groupby("pa_id")["pitch_type"].shift(-1)

data_train = data[data["target_is_real_pitch"] == True].copy()
```

In [13]:
```
# randomly select plate appearances to be a part of training and test sets

def split_by_pa_id(df: pd.DataFrame, pa_col="pa_id", ratios=(0.8, 0.2), seed: int=4
    r_train, r_test = ratios
    assert abs((r_train + r_test) - 1.0) < 1e-9

    pa_ids = df[pa_col].dropna().unique()

    rng = np.random.default_rng(seed)
    rng.shuffle(pa_ids)

    n = len(pa_ids)
    n_train = int(n*r_train)

    train_ids = set(pa_ids[:n_train])
    test_ids = set(pa_ids[n_train:])

    train_df = df[df[pa_col].isin(train_ids)].copy()
    test_df = df[df[pa_col].isin(test_ids)].copy()

    return train_df, test_df, train_ids, test_ids
```

In [14]:
```
train_df, test_df, train_ids, test_ids = split_by_pa_id(
    data_train, pa_col="pa_id", ratios=(0.8, 0.2), seed=7
)
```

```
C:\Users\sethb\AppData\Local\Temp\ipykernel_31780\1463695035.py:10: UserWarning: you
are shuffling a 'StringArray' object which is not a subclass of 'Sequence'; `shuffle
` is not guaranteed to behave correctly. E.g., non-numpy array/tensor objects with v
iew semantics may contain duplicates after shuffling.
  rng.shuffle(pa_ids)
```

In [15]:
```
# Features to be used by the model
FEATURE_SPEC = {
    "target": "y_next_pitch_type",
    "cat_cols": [
        "pitcher", "batter", "stand", "p_throws", "inning_topbot",
        "count_state", "prev_pitch_type"
    ],
    "num_cols": [
        "balls", "strikes", "outs_when_up", "inning", "score_diff_bat",
        "on_1b", "on_2b", "on_3b"
    ],
}

TARGET_COL = FEATURE_SPEC["target"]
CAT_COLS = FEATURE_SPEC["cat_cols"]
NUM_COLS = FEATURE_SPEC["num_cols"]
```

Numeric Scaling

I added a StandardScaler here which will be applied to all numeric columns. This will be more impactful as features with more variance are added, given that te current set of numeric features don't have many possible values.

In [16]:
```
PAD_ID = 0

def build_vocab(values):
    uniq = pd.Series(values.dropna().unique())
    return {v: i for i, v in enumerate(uniq, start=1)}

def encode(series, vocab):
    return series.map(vocab).fillna(PAD_ID).astype(int)

cat_vocabs = {c: build_vocab(train_df[c]) for c in CAT_COLS}
y_vocab    = build_vocab(train_df[TARGET_COL])
cat_vocab_sizes = {c: len(cat_vocabs[c]) + 1 for c in CAT_COLS}
num_classes = len(y_vocab) + 1

scaler = StandardScaler()
scaler.fit(train_df[NUM_COLS].fillna(0))

def encode_df(df):
    out = df.copy()
    for c in CAT_COLS:
        out[c + "_id"] = encode(out[c], cat_vocabs[c])
    out["y_id"] = encode(out[TARGET_COL], y_vocab)
    for c in NUM_COLS:
        out[c] = pd.to_numeric(out[c], errors="coerce").fillna(0).astype(np.float32

    out[NUM_COLS] = scaler.transform(out[NUM_COLS])
    return out

train_enc = encode_df(train_df)
test_enc  = encode_df(test_df)
```

In [17]:
```python
# make all pitch sequences the same length
def make_fixed_sequences(df, pa_col="pa_id", max_len=8):
    X_cat, X_num, Y = [], [], []

    for _, g in df.groupby(pa_col, sort=False):

        cat = g[[c + "_id" for c in CAT_COLS]].to_numpy(np.int64)
        num = g[NUM_COLS].to_numpy(np.float32)
        y   = g["y_id"].to_numpy(np.int64)

        L = min(len(g), max_len)
        cat, num, y = cat[:L], num[:L], y[:L]

        pad = max_len - L
        if pad > 0:
            cat = np.pad(cat, ((0,pad),(0,0)), constant_values=PAD_ID)
            num = np.pad(num, ((0,pad),(0,0)), constant_values=0.0)
            y   = np.pad(y,   (0,pad),         constant_values=PAD_ID)

        X_cat.append(cat); X_num.append(num); Y.append(y)

    return (
        torch.tensor(np.stack(X_cat), dtype=torch.long),
        torch.tensor(np.stack(X_num), dtype=torch.float32),
        torch.tensor(np.stack(Y),     dtype=torch.long),
    )

MAX_LEN = 8
Xc_tr, Xn_tr, Y_tr = make_fixed_sequences(train_enc, max_len=MAX_LEN)
Xc_te, Xn_te, Y_te = make_fixed_sequences(test_enc,  max_len=MAX_LEN)
```

In [18]:
```python
# Create the Dataset
class PitchSeqDS(Dataset):
    def __init__(self, Xc, Xn, Y):
        self.Xc, self.Xn, self.Y = Xc, Xn, Y
    def __len__(self): return self.Y.size(0)
    def __getitem__(self, i): return self.Xc[i], self.Xn[i], self.Y[i]

train_loader = DataLoader(PitchSeqDS(Xc_tr, Xn_tr, Y_tr), batch_size=64, shuffle=Tr
test_loader  = DataLoader(PitchSeqDS(Xc_te, Xn_te, Y_te), batch_size=64, shuffle=Fa
```

In [19]:
```python
from datetime import datetime
from pathlib import Path
import pandas as pd

# Export vocabularies and feature lists with a date-stamped filename.
today = datetime.now().strftime("%Y%m%d")
repo_root = Path.cwd().parent
vocab_dir = repo_root / "model_shared" / "vocab"
feature_dir = repo_root / "model_shared" / "feature-list"
vocab_dir.mkdir(parents=True, exist_ok=True)
feature_dir.mkdir(parents=True, exist_ok=True)

rows = []
for feature, vocab in cat_vocabs.items():
```

```
    for value, idx in vocab.items():
        rows.append({"feature": feature, "value": value, "id": idx, "kind": "catego
for value, idx in y_vocab.items():
    rows.append({"feature": TARGET_COL, "value": value, "id": idx, "kind": "target"

vocab_path = vocab_dir / f"rnn_vocab_{today}.csv"
pd.DataFrame(rows).to_csv(vocab_path, index=False)

feature_rows = []
for c in CAT_COLS:
    feature_rows.append({"feature": c, "kind": "categorical"})
for c in NUM_COLS:
    feature_rows.append({"feature": c, "kind": "numerical"})
feature_rows.append({"feature": TARGET_COL, "kind": "target"})

feature_path = feature_dir / f"rnn_vocab_{today}.csv"
pd.DataFrame(feature_rows).to_csv(feature_path, index=False)

print(f"Wrote vocab to {vocab_path}")
print(f"Wrote feature list to {feature_path}")
```

```
Wrote vocab to c:\Users\sethb\Baseball\pitchcraft-model\model_shared\vocab\rnn_vocab
_20260214.csv
Wrote feature list to c:\Users\sethb\Baseball\pitchcraft-model\model_shared\feature-
list\rnn_vocab_20260214.csv
```

Model Definitions

For this model version, I opted to create new model definitions with just one change from the original model definition in V1. The SimplePitchRNN now contains code for handling custom embedding sizes, and the PitchRNNWithDropout initializes a dropout layer. These two models will be combined in V3 and all subsequent versions that do not have changes to how the model is defined.

In [20]:
```
# Model Simple RNN Model
class SimplePitchRNN(nn.Module):
    def __init__(self, cat_vocab_sizes, num_features, emb_dims=None, hidden=128, nu
        super().__init__()
        self.cat_cols = list(cat_vocab_sizes.keys())

        if emb_dims is None:
            emb_dims = {col: 16 for col in self.cat_cols}

        self.embs = nn.ModuleDict({
            col: nn.Embedding(cat_vocab_sizes[col], emb_dims[col], padding_idx=pad_
            for col in self.cat_cols
        })

        in_dim = sum(emb_dims.values())+ num_features
        self.rnn = nn.RNN(in_dim, hidden, batch_first=True)
        self.fc  = nn.Linear(hidden, num_classes)

    def forward(self, x_cat, x_num):
        embs = []
```

```
        for j, col in enumerate(self.cat_cols):
            embs.append(self.embs[col](x_cat[:, :, j]))
        x = torch.cat(embs + [x_num], dim=-1)
        h, _ = self.rnn(x)
        return self.fc(h)
```

In [21]:
```
# Model with Dropout
class PitchRNNWithDropout(nn.Module):
    def __init__(self, cat_vocab_sizes, num_features, emb_dim=16, hidden=128, num_c
        super().__init__()
        self.cat_cols = list(cat_vocab_sizes.keys())

        self.embs = nn.ModuleDict({
            col: nn.Embedding(cat_vocab_sizes[col], emb_dim, padding_idx=pad_id)
            for col in self.cat_cols
        })

        self.emb_dropout = nn.Dropout(dropout)

        in_dim = len(self.cat_cols) * emb_dim + num_features
        self.rnn = nn.RNN(in_dim, hidden, batch_first=True, dropout=dropout if drop
        self.fc_dropout = nn.Dropout(dropout)
        self.fc  = nn.Linear(hidden, num_classes)

    def forward(self, x_cat, x_num):
        embs = []
        for j, col in enumerate(self.cat_cols):
            embs.append(self.embs[col](x_cat[:, :, j]))

        embs = [self.emb_dropout(emb) for emb in embs]

        x = torch.cat(embs + [x_num], dim=-1)
        h, _ = self.rnn(x)
        h = self.fc_dropout(h)
        return self.fc(h)
```

Model improvements

These improvements (class weights, early stopping) are new features that were added this version to help improve the performance of the model. The results of these improvements can be found later on in the notebook.

The calculate class weights function computes the inverse frequency weights for each pitch class, with a smoothing exponent of 0.2. This prevents the model from being too overly punished for rare pitch types, with the idea that fastballs will not be so heavily predicted. The output of this function is a Tensor which is used in the CrossEntropyLoss evaluation function later on.

In [22]:
```
# Calculating Class Weights
def calculate_class_weights(y_train, pad_id=0, smoothing=0.2):
    y_flat = y_train.flatten()
    y_flat = y_flat[y_flat != pad_id]
```

```
    unique, counts = np.unique(y_flat, return_counts=True)

    total = len(y_flat)
    frequencies = counts / total
    weights = (1.0 / frequencies) ** smoothing

    weight_tensor = torch.zeros(num_classes)
    for class_id, weight in zip (unique, weights):
        weight_tensor[class_id] = weight
    return weight_tensor

class_weights = calculate_class_weights(Y_tr)
class_weights
```

Out[22]:
```
tensor([0.0000, 1.4619, 1.2596, 1.9154, 1.4753, 1.6947, 1.7173, 2.2490, 1.6944,
        1.5371, 3.8652, 2.8790, 4.2934, 3.8620, 4.0726])
```

The early stopping class tracks validation loss across epochs, saves the best model state, and checks to stop when the loss fails to improve by at least the value delta over patience consecutive epochs. This class also provides a load_best_model method that restores the best checkpoint at th end of training. The purpose of this class is to prevent the model from overfitting during the training process.

In [23]:
```
# Creating a class to track early stopping
class EarlyStopping:
    def __init__(self, patience=5, delta=0):
        self.patience = patience
        self.delta = delta
        self.best_score = None
        self.early_stop = False
        self.counter = 0
        self.best_model_state = None

    def __call__(self, val_loss, model):
        score = -val_loss

        if self.best_score is None:
            self.best_score = score
            self.best_model_state = model.state_dict()
        elif score < self.best_score + self.delta:
            self.counter += 1
            if self.counter >= self.patience:
                self.early_stop = True
        else:
            self.best_score = score
            self.best_model_state = model.state_dict()
            self.counter = 0

    def load_best_model(self, model):
        model.load_state_dict(self.best_model_state)
```

Training Functions

For V2 I tweaked how the training works slightly. I took the original training code and put it in a function train_with_weights() so that I could have some control over how the model would go about training. The code inside the function is identical to V1, with the only change being to how I start the training process.

In [24]:
```
# Simplified the training loop
def train_with_weights(model, criterion, optimizer, device):
    epochs = 20
    for epoch in range(epochs):
        model.train()
        epoch_loss = 0
        for x_cat, x_num, y in train_loader:
            x_cat = x_cat.to(device)
            x_num = x_num.to(device)
            y     = y.to(device)
            logits = model(x_cat, x_num)

            loss = criterion(
                logits.reshape(-1, num_classes),
                y.reshape(-1)
            )

            optimizer.zero_grad()
            loss.backward()
            optimizer.step()

            epoch_loss += loss.item()

        print(f'Epoch [{epoch+1}/{epochs}], Loss: {epoch_loss / len(train_loader):.
```

Then when I implemented early stopping, I added calculation for the validation loss to the original training loop. Moving forward, this is going to be the training function we use so we can keep track of when to stop the model.

In [25]:
```
def train_with_early_stopping(model, train_loader, test_loader, criterion, optimize
    epochs = 20
    for epoch in range(epochs):
        model.train()
        train_loss = 0
        for x_cat, x_num, y in train_loader:
            x_cat = x_cat.to(device)
            x_num = x_num.to(device)
            y     = y.to(device)
            logits = model(x_cat, x_num)

            loss = criterion(
                logits.reshape(-1, num_classes),
                y.reshape(-1)
            )

            optimizer.zero_grad()
            loss.backward()
            optimizer.step()
```

```
            train_loss += loss.item() * x_cat.size(0)

        train_loss /= len(train_loader.dataset)

        model.eval()
        test_loss = 0
        with torch.no_grad():
            for x_cat, x_num, y in test_loader:
                x_cat = x_cat.to(device)
                x_num = x_num.to(device)
                y     = y.to(device)

                logits = model(x_cat, x_num)
                loss = criterion(
                    logits.reshape(-1, num_classes),
                    y.reshape(-1)
                )
                test_loss += loss.item() * x_cat.size(0)

        test_loss /= len(test_loader.dataset)

        print(f'Epoch {epoch+1}, Train Loss: {train_loss:.4f}, Test Loss: {test_los

        early_stopping(test_loss, model)
        if early_stopping.early_stop:
            print("Early Stopping")
            break

    early_stopping.load_best_model(model)
```

Execute Training

Here is where some of the reorganization starts to pay off. For each model change implemented, there is a cell which sets up these variables for the model, criterion, and optimizer. Eventually these can be put into configuration files so that we can easily tweak the parameters we train on, but this was just the first step towards that.

In [ ]:
```
# Model Training using weights
device = torch.device("cuda" if torch.cuda.is_available() else "cpu")
weights_model = SimplePitchRNN(cat_vocab_sizes, num_features=len(NUM_COLS), num_cla
weights_model = weights_model.to(device)
criterion = nn.CrossEntropyLoss(ignore_index=PAD_ID, weight=class_weights)
optimizer = optim.Adam(weights_model.parameters(), lr=0.001)

train_with_weights(weights_model, criterion, optimizer, device)
```

```
Epoch [1/20], Loss: 1.7441
Epoch [2/20], Loss: 1.4907
Epoch [3/20], Loss: 1.3957
Epoch [4/20], Loss: 1.3506
Epoch [5/20], Loss: 1.3224
Epoch [6/20], Loss: 1.3048
Epoch [7/20], Loss: 1.2918
Epoch [8/20], Loss: 1.2819
Epoch [9/20], Loss: 1.2734
Epoch [10/20], Loss: 1.2668
Epoch [11/20], Loss: 1.2605
Epoch [12/20], Loss: 1.2550
Epoch [13/20], Loss: 1.2505
Epoch [14/20], Loss: 1.2459
Epoch [15/20], Loss: 1.2416
Epoch [16/20], Loss: 1.2374
Epoch [17/20], Loss: 1.2337
Epoch [18/20], Loss: 1.2306
Epoch [19/20], Loss: 1.2266
Epoch [20/20], Loss: 1.2237
```

In [ ]:
```python
# Model with Early Stopping
early_stopping_model = SimplePitchRNN(cat_vocab_sizes, num_features=len(NUM_COLS),
early_stopping_model = early_stopping_model.to(device)
criterion = nn.CrossEntropyLoss(ignore_index=PAD_ID)
optimizer = optim.Adam(early_stopping_model.parameters(), lr=0.001)
early_stopping = EarlyStopping(patience=5, delta=0.01)

train_with_early_stopping(early_stopping_model, train_loader, test_loader, criterio
```

```
Epoch 1, Train Loss: 1.6888, Test Loss: 1.5372
Epoch 2, Train Loss: 1.4554, Test Loss: 1.4050
Epoch 3, Train Loss: 1.3641, Test Loss: 1.3514
Epoch 4, Train Loss: 1.3221, Test Loss: 1.3251
Epoch 5, Train Loss: 1.2978, Test Loss: 1.3146
Epoch 6, Train Loss: 1.2818, Test Loss: 1.3018
Epoch 7, Train Loss: 1.2707, Test Loss: 1.2988
Epoch 8, Train Loss: 1.2619, Test Loss: 1.2926
Epoch 9, Train Loss: 1.2544, Test Loss: 1.2883
Epoch 10, Train Loss: 1.2483, Test Loss: 1.2890
Epoch 11, Train Loss: 1.2428, Test Loss: 1.2920
Epoch 12, Train Loss: 1.2375, Test Loss: 1.2846
Epoch 13, Train Loss: 1.2337, Test Loss: 1.2839
Epoch 14, Train Loss: 1.2293, Test Loss: 1.2842
Early Stopping
```

In [ ]:
```python
# Model with dropout layer and early stopping and class weights
dropout_model = PitchRNNWithDropout(
    cat_vocab_sizes,
    num_features=len(NUM_COLS),
    num_classes=num_classes,
    dropout=0.1
)
dropout_model = dropout_model.to(device)

criterion = nn.CrossEntropyLoss(ignore_index=PAD_ID, weight=class_weights)
```

```
optimizer = optim.Adam(dropout_model.parameters(), lr=0.001)
early_stopping = EarlyStopping(patience=5, delta=0.01)

train_with_early_stopping(dropout_model, train_loader, test_loader,
                          criterion, optimizer, early_stopping, device)
```

```
c:\Users\sethb\AppData\Local\Python\pythoncore-3.11-64\Lib\site-packages\torch\nn\modules\rnn.py:641: UserWarning: dropout option adds dropout after all but last recurrent layer, so non-zero dropout expects num_layers greater than 1, but got dropout=0.1 and num_layers=1
  super().__init__(mode, *args, **kwargs)
Epoch 1, Train Loss: 1.7842, Test Loss: 1.5957
Epoch 2, Train Loss: 1.5532, Test Loss: 1.4633
Epoch 3, Train Loss: 1.4621, Test Loss: 1.3995
Epoch 4, Train Loss: 1.4149, Test Loss: 1.3705
Epoch 5, Train Loss: 1.3883, Test Loss: 1.3491
Epoch 6, Train Loss: 1.3688, Test Loss: 1.3356
Epoch 7, Train Loss: 1.3550, Test Loss: 1.3272
Epoch 8, Train Loss: 1.3454, Test Loss: 1.3183
Epoch 9, Train Loss: 1.3362, Test Loss: 1.3129
Epoch 10, Train Loss: 1.3298, Test Loss: 1.3100
Epoch 11, Train Loss: 1.3252, Test Loss: 1.3069
Epoch 12, Train Loss: 1.3192, Test Loss: 1.3043
Epoch 13, Train Loss: 1.3160, Test Loss: 1.3010
Epoch 14, Train Loss: 1.3122, Test Loss: 1.3049
Epoch 15, Train Loss: 1.3088, Test Loss: 1.3010
Epoch 16, Train Loss: 1.3063, Test Loss: 1.2991
Early Stopping
```

In [ ]:

```
# Model with custom embedding dims, class weights, and early stopping
custom_emb_dims = {
    "pitcher": 32,
    "batter": 32,
    "stand": 2,
    "p_throws": 2,
    "inning_topbot": 2,
    "count_state": 12,
    "prev_pitch_type": 16
}

emb_model = SimplePitchRNN(
    cat_vocab_sizes=cat_vocab_sizes,
    num_features=len(NUM_COLS),
    emb_dims=custom_emb_dims,
    num_classes=num_classes
)

emb_model = emb_model.to(device)

criterion = nn.CrossEntropyLoss(ignore_index=PAD_ID, weight=class_weights)
optimizer = optim.Adam(emb_model.parameters(), lr=0.001)
early_stopping = EarlyStopping(patience=5, delta=0.01)

train_with_early_stopping(emb_model, train_loader, test_loader,
                          criterion, optimizer, early_stopping, device)
```

```
Epoch 1, Train Loss: 1.6847, Test Loss: 1.4878
Epoch 2, Train Loss: 1.4083, Test Loss: 1.3729
Epoch 3, Train Loss: 1.3372, Test Loss: 1.3399
Epoch 4, Train Loss: 1.3059, Test Loss: 1.3207
Epoch 5, Train Loss: 1.2873, Test Loss: 1.3158
Epoch 6, Train Loss: 1.2743, Test Loss: 1.3110
Epoch 7, Train Loss: 1.2641, Test Loss: 1.3050
Epoch 8, Train Loss: 1.2553, Test Loss: 1.3084
Epoch 9, Train Loss: 1.2476, Test Loss: 1.3051
Epoch 10, Train Loss: 1.2411, Test Loss: 1.3064
Epoch 11, Train Loss: 1.2349, Test Loss: 1.3083
Epoch 12, Train Loss: 1.2288, Test Loss: 1.3105
Early Stopping
```

Evaluation

For this model version I also reconfigured the evaluation portion to be more in line with our design document. The following cell acts as a class which contains all the functions needed to evaluate the model. The code for these functions is the same as V1, just organized into one spot.

In [ ]:

```python
from sklearn.metrics import classification_report
from sklearn.metrics import confusion_matrix
from collections import Counter

def get_all_predictions(model, test_loader, device, pad_id = 0):
    model.eval()
    all_preds = []
    all_true = []

    with torch.no_grad():
        for x_cat, x_num, y in test_loader:
            x_cat = x_cat.to(device)
            x_num = x_num.to(device)
            y = y.to(device)
            logits = model(x_cat, x_num)        # (B, L, C)
            preds  = logits.argmax(dim=-1)      # (B, L)

            mask = (y != pad_id)

            valid = mask.bool()

            all_preds.append(preds[valid].cpu().numpy())
            all_true.append(y[valid].cpu().numpy())

    y_pred = np.concatenate(all_preds)
    y_true = np.concatenate(all_true)

    return y_true, y_pred

def print_classification_report(model, test_loader, device, id_to_pitch, pad_id=0):
    y_true, y_pred = get_all_predictions(model, test_loader, device, pad_id)
    target_names = list(id_to_pitch.values())
    report = classification_report(y_true, y_pred, target_names=target_names)
```

```
    print(report)
    return report

def get_strategic_accuracy(model, test_loader, device, id_to_pitch, pad_id=0):
    y_true, y_pred = get_all_predictions(model, test_loader, device, pad_id)

    fastballs = {'FF', 'SI', 'FC'}
    breaking = {'SL', 'CU', 'KC', 'SV', 'ST'}
    offspeed = {'CH', 'FS'}

    def pitch_to_family(pitch):
        if pitch in fastballs: return 'fastball'
        if pitch in breaking: return 'breaking'
        if pitch in offspeed: return 'offspeed'
        return 'other'

    y_true_family = [pitch_to_family(id_to_pitch[y]) for y in y_true]
    y_pred_family = [pitch_to_family(id_to_pitch[p]) for p in y_pred]

    report = classification_report(y_true_family, y_pred_family)
    print(report)
    return report

def generate_confusion_matrix(model, test_loader, device, pad_id=0, figsize=(12, 10
    """Generate and display confusion matrix."""
    y_true, y_pred = get_all_predictions(model, test_loader, device, pad_id)

    cm = confusion_matrix(y_true, y_pred)
    cm_norm = cm / cm.sum(axis=1, keepdims=True)

    plt.figure(figsize=figsize)
    sns.heatmap(cm_norm, cmap="Blues", fmt="d")
    plt.xlabel("Predicted Pitch Type")
    plt.ylabel("True Pitch Type")
    plt.title("Pitch Type Confusion Matrix (Token-Level)")
    plt.show()

    return cm, cm_norm

def get_accuracy(model, test_loader, device, pad_id=0):
    y_true, y_pred = get_all_predictions(model, test_loader, device, pad_id)
    correct = (y_pred == y_true).sum()
    total = len(y_true)
    accuracy = 100 * correct / total
    print(f"Token Accuracy (no PAD): {accuracy:.2f}%")
    return accuracy

def get_top_k_accuracy(model, test_loader, device, K=3, pad_id=0):
    model.eval()
    correct = 0
    total = 0

    with torch.no_grad():
        for x_cat, x_num, y in test_loader:
            x_cat = x_cat.to(device)
            x_num = x_num.to(device)
```

```
            y = y.to(device)

            logits = model(x_cat, x_num)
            topk = logits.topk(K, dim=-1).indices
            mask = (y != pad_id)
            match = (topk == y.unsqueeze(-1)).any(dim=-1)
            correct += (match & mask).sum().item()
            total += mask.sum().item()

    accuracy = 100 * correct / total
    print(f"Top-{K} Accuracy: {accuracy:.2f}%")
    return accuracy

def get_most_common_pitches(model, test_loader, device, id_to_pitch, top_n=5, pad_i
    model.eval()
    counts = Counter()

    with torch.no_grad():
        for x_cat, x_num, y in test_loader:
            x_cat = x_cat.to(device)
            x_num = x_num.to(device)
            preds = model(x_cat, x_num).argmax(dim=-1)
            for p in preds.view(-1).tolist():
                if p != pad_id:
                    counts[p] += 1

    print(f"Top {top_n} most predicted pitches:")
    for pid, cnt in counts.most_common(top_n):
        print(f"  {id_to_pitch.get(pid, 'PAD')}: {cnt}")

    return counts.most_common(top_n)
```

Then to evaluate the models, I can generate a report in one output instead of across several cells in V1. This helps to keep track of what metrics belong to which models

In [ ]:
```
def evaluate_model_complete(model, test_loader, device, id_to_pitch, num_classes, p

    print("\n1. Token Accuracy:")
    acc = get_accuracy(model, test_loader, device, pad_id)

    print("\n2. Top-3 Accuracy:")
    top3 = get_top_k_accuracy(model, test_loader, device, K=3, pad_id=pad_id)

    print("\n3. Most Common Predictions:")
    common = get_most_common_pitches(model, test_loader, device, id_to_pitch, pad_i

    print("\n4. Detailed Classification Report:")
    class_report = print_classification_report(model, test_loader, device, id_to_pi

    print("\n5. Strategic Accuracy (by pitch family):")
    strat_report = get_strategic_accuracy(model, test_loader, device, id_to_pitch,

    print("\n6. Confusion Matrix:")
    cm, cm_norm = generate_confusion_matrix(model, test_loader, device, pad_id)
```

```
    return {
        'accuracy': acc,
        'top3_accuracy': top3,
        'most_common': common,
        'classification_report': class_report,
        'strategic_report': strat_report,
        'confusion_matrix': cm,
        'confusion_matrix_normalized': cm_norm
    }
```

Now all it takes to evaluate the mode is this chunk of code where you pass in the model use, the test DataLoader, which device to run, the mapping of ids to pitch types from the encoder, the total number of classes, and lastly the PAD_ID

In [ ]:
```
id_to_pitch = {v: k for k, v in y_vocab.items()}

weights_results = evaluate_model_complete(
    model=weights_model,
    test_loader=test_loader,
    device=device,
    id_to_pitch=id_to_pitch,
    num_classes=num_classes,
    pad_id=PAD_ID
)
```

1. Token Accuracy:
Token Accuracy (no PAD): 42.11%

2. Top-3 Accuracy:
Top-3 Accuracy: 87.37%

3. Most Common Predictions:
Top 5 most predicted pitches:
  SL: 57958
  FF: 54108
  SI: 43953
  CH: 28291
  ST: 25980

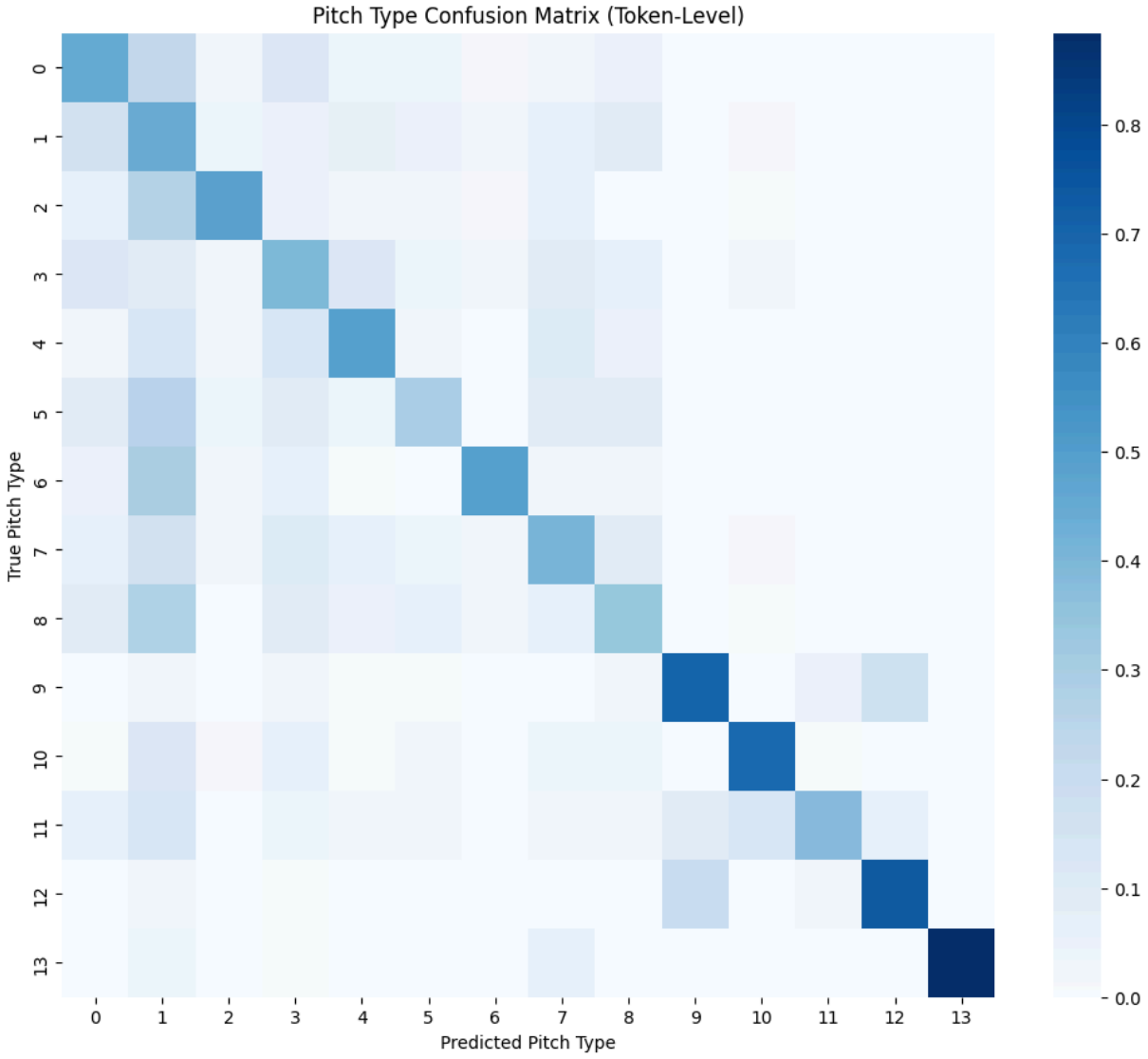
4. Detailed Classification Report:

	precision	recall	f1-score	support
SL	0.43	0.45	0.44	15820
FF	0.51	0.44	0.47	33475
FS	0.40	0.48	0.44	3889
SI	0.44	0.40	0.42	15342
ST	0.36	0.49	0.42	7596
CU	0.33	0.29	0.31	7114
KC	0.34	0.50	0.41	1781
FC	0.32	0.41	0.36	7658
CH	0.39	0.34	0.36	12150
FA	0.76	0.70	0.73	142
SV	0.26	0.68	0.37	530
OTHER	0.28	0.38	0.32	71
EP	0.71	0.74	0.72	103
FO	0.40	0.88	0.55	112
accuracy			0.42	105783
macro avg	0.42	0.51	0.45	105783
weighted avg	0.43	0.42	0.42	105783

5. Strategic Accuracy (by pitch family):

	precision	recall	f1-score	support
breaking	0.46	0.53	0.49	32841
fastball	0.62	0.58	0.60	56475
offspeed	0.39	0.38	0.38	16039
other	0.63	0.86	0.73	428
accuracy			0.53	105783
macro avg	0.53	0.59	0.55	105783
weighted avg	0.54	0.53	0.53	105783

6. Confusion Matrix:



In [ ]:
```
early_stopping_results = evaluate_model_complete(
    model=early_stopping_model,
    test_loader=test_loader,
    device=device,
    id_to_pitch=id_to_pitch,
    num_classes=num_classes,
    pad_id=PAD_ID
)
```

1. Token Accuracy:
Token Accuracy (no PAD): 43.88%

2. Top-3 Accuracy:
Top-3 Accuracy: 87.79%

3. Most Common Predictions:
Top 5 most predicted pitches:
  FF: 106518
  SI: 55555
  SL: 28685
  CH: 22448
  CU: 18732

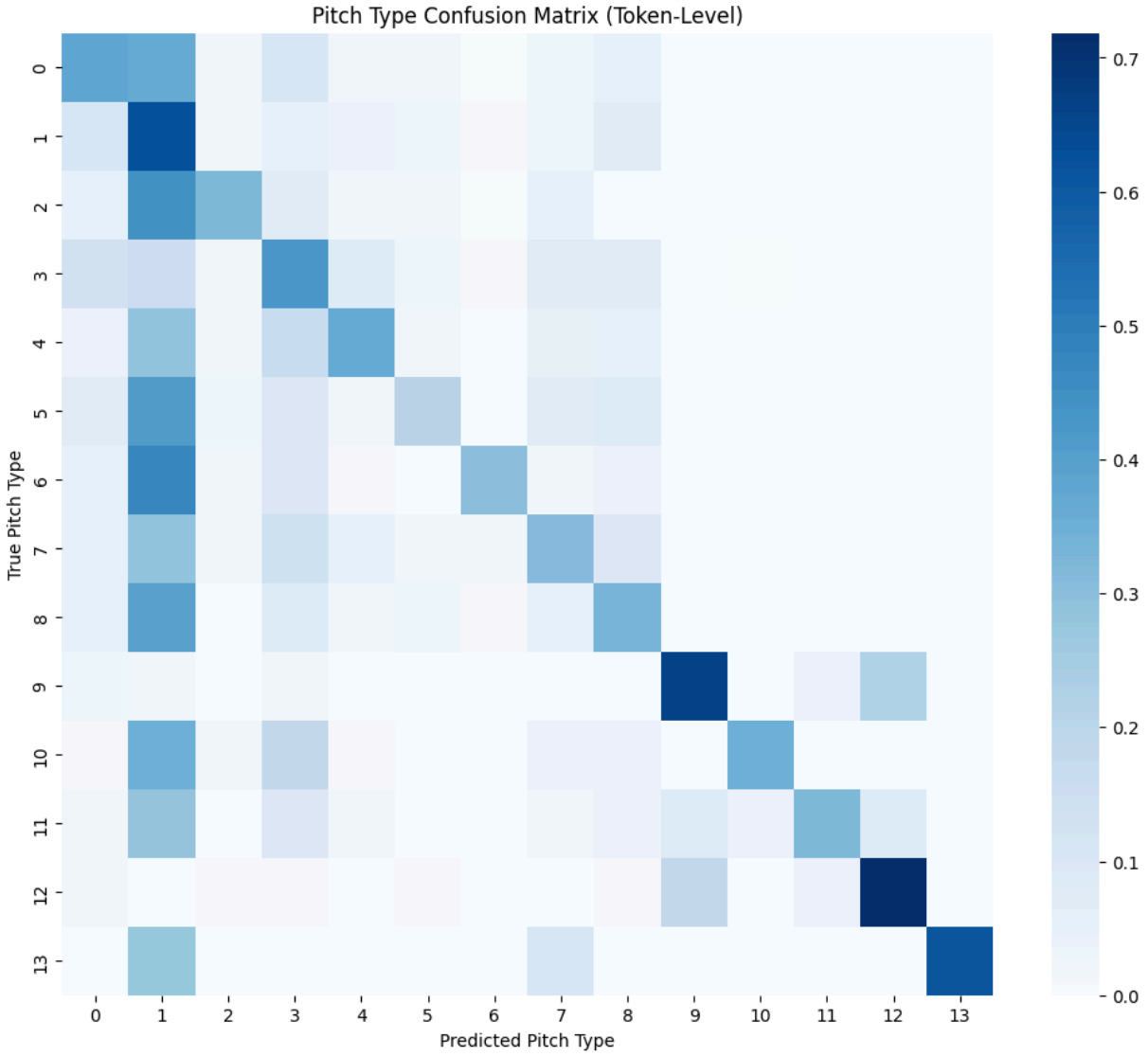
4. Detailed Classification Report:

	precision	recall	f1-score	support
SL	0.44	0.38	0.41	15820
FF	0.48	0.63	0.54	33475
FS	0.42	0.32	0.37	3889
SI	0.45	0.42	0.43	15342
ST	0.41	0.37	0.39	7596
CU	0.37	0.21	0.27	7114
KC	0.38	0.30	0.34	1781
FC	0.35	0.31	0.33	7658
CH	0.39	0.33	0.36	12150
FA	0.77	0.66	0.71	142
SV	0.33	0.35	0.34	530
OTHER	0.51	0.32	0.40	71
EP	0.64	0.72	0.68	103
FO	0.49	0.62	0.55	112
accuracy			0.44	105783
macro avg	0.46	0.42	0.44	105783
weighted avg	0.43	0.44	0.43	105783

5. Strategic Accuracy (by pitch family):

	precision	recall	f1-score	support
breaking	0.49	0.39	0.44	32841
fastball	0.60	0.70	0.64	56475
offspeed	0.40	0.33	0.36	16039
other	0.79	0.78	0.78	428
accuracy			0.55	105783
macro avg	0.57	0.55	0.56	105783
weighted avg	0.54	0.55	0.54	105783

6. Confusion Matrix:



In [ ]:
```
dropout_results = evaluate_model_complete(
    model=dropout_model,
    test_loader=test_loader,
    device=device,
    id_to_pitch=id_to_pitch,
    num_classes=num_classes,
    pad_id=PAD_ID
)
```

1. Token Accuracy:
Token Accuracy (no PAD): 42.96%

2. Top-3 Accuracy:
Top-3 Accuracy: 87.73%

3. Most Common Predictions:
Top 5 most predicted pitches:
  FF: 138525
  SI: 44324
  CH: 17004
  SL: 15621
  ST: 14223

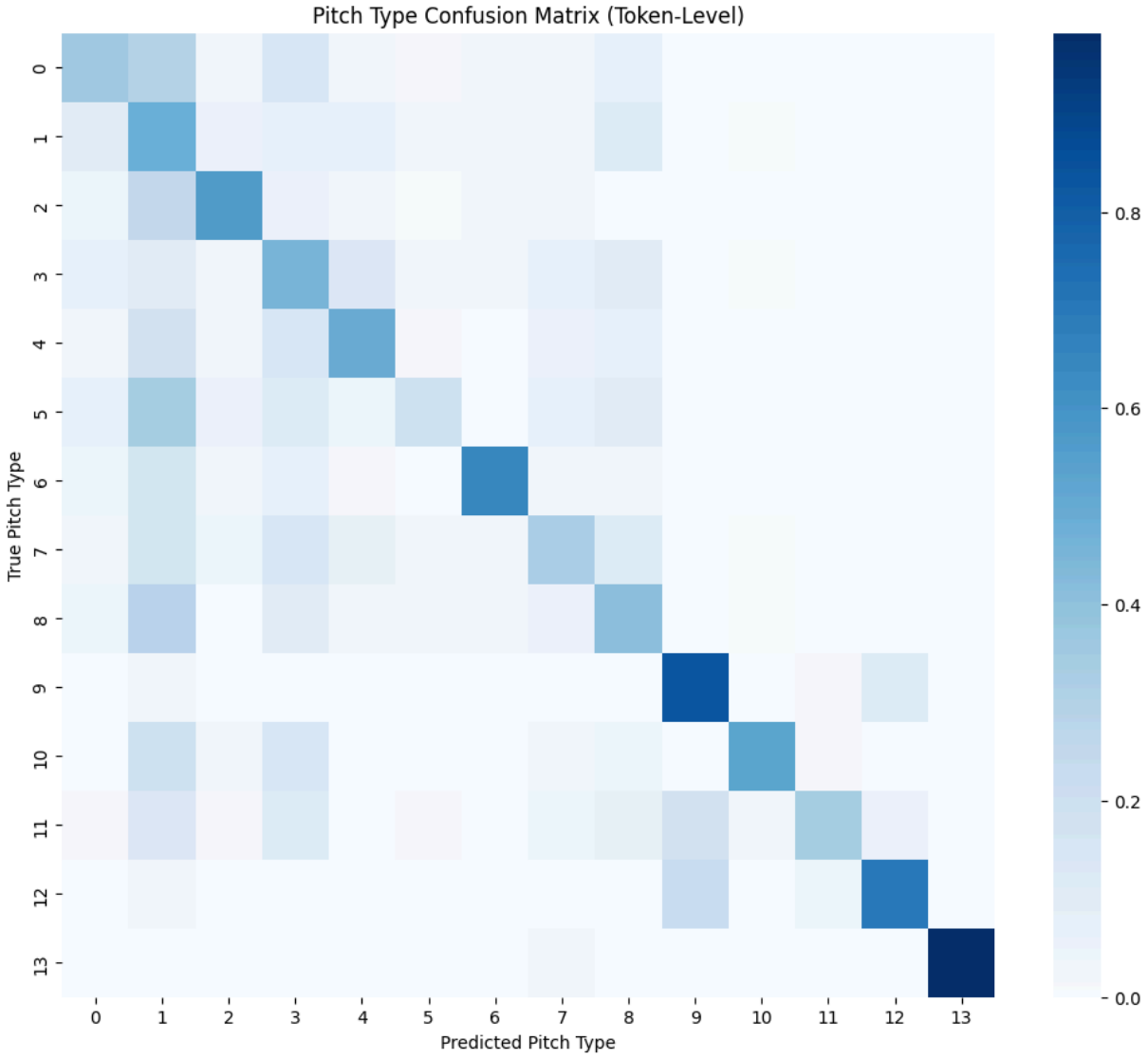
4. Detailed Classification Report:

	precision	recall	f1-score	support
SL	0.47	0.35	0.40	15820
FF	0.50	0.49	0.49	33475
FS	0.39	0.56	0.46	3889
SI	0.43	0.46	0.44	15342
ST	0.37	0.49	0.42	7596
CU	0.38	0.20	0.26	7114
KC	0.31	0.65	0.42	1781
FC	0.35	0.32	0.33	7658
CH	0.37	0.41	0.39	12150
FA	0.75	0.84	0.79	142
SV	0.28	0.53	0.37	530
OTHER	0.20	0.34	0.25	71
EP	0.77	0.70	0.73	103
FO	0.38	0.98	0.55	112
accuracy			0.43	105783
macro avg	0.43	0.52	0.45	105783
weighted avg	0.44	0.43	0.43	105783

5. Strategic Accuracy (by pitch family):

	precision	recall	f1-score	support
breaking	0.47	0.44	0.46	32841
fastball	0.61	0.60	0.61	56475
offspeed	0.38	0.45	0.41	16039
other	0.58	0.91	0.71	428
accuracy			0.53	105783
macro avg	0.51	0.60	0.55	105783
weighted avg	0.53	0.53	0.53	105783

6. Confusion Matrix:



In [ ]:
```
emb_results = evaluate_model_complete(
    model=emb_model,
    test_loader=test_loader,
    device=device,
    id_to_pitch=id_to_pitch,
    num_classes=num_classes,
    pad_id=PAD_ID
)
```

1. Token Accuracy:
Token Accuracy (no PAD): 42.43%

2. Top-3 Accuracy:
Top-3 Accuracy: 87.55%

3. Most Common Predictions:
Top 5 most predicted pitches:
  FF: 104531
  SI: 80098
  SL: 25403
  CH: 18035
  FC: 8909

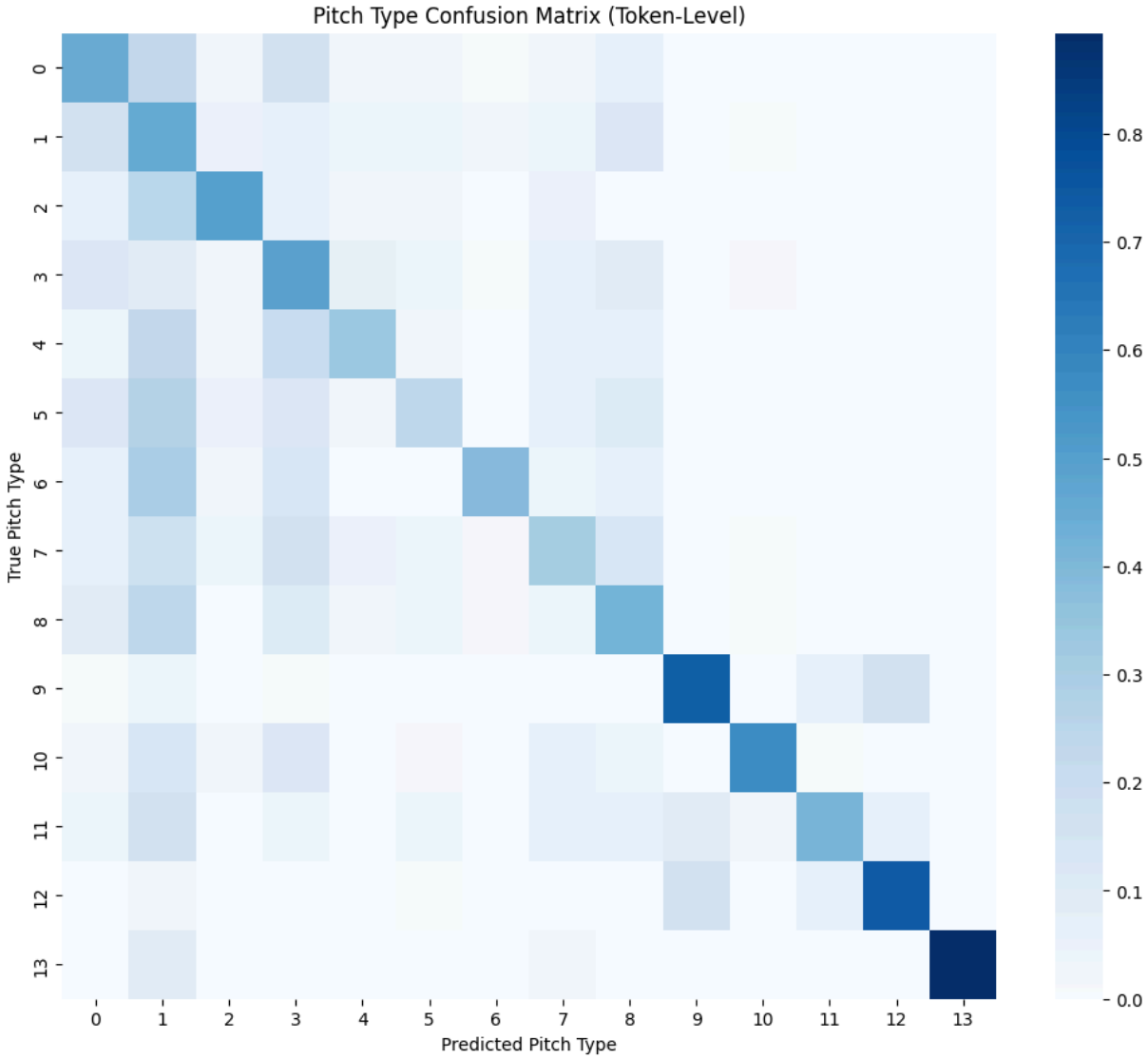
4. Detailed Classification Report:

	precision	recall	f1-score	support
SL	0.42	0.44	0.43	15820
FF	0.51	0.45	0.48	33475
FS	0.39	0.50	0.44	3889
SI	0.42	0.49	0.45	15342
ST	0.40	0.34	0.37	7596
CU	0.33	0.24	0.28	7114
KC	0.37	0.38	0.38	1781
FC	0.35	0.31	0.33	7658
CH	0.36	0.43	0.39	12150
FA	0.81	0.73	0.77	142
SV	0.29	0.57	0.39	530
OTHER	0.34	0.41	0.37	71
EP	0.72	0.74	0.73	103
FO	0.41	0.89	0.56	112
accuracy			0.42	105783
macro avg	0.44	0.50	0.45	105783
weighted avg	0.43	0.42	0.42	105783

5. Strategic Accuracy (by pitch family):

	precision	recall	f1-score	support
breaking	0.47	0.45	0.46	32841
fastball	0.62	0.60	0.61	56475
offspeed	0.37	0.45	0.41	16039
other	0.66	0.87	0.75	428
accuracy			0.53	105783
macro avg	0.53	0.59	0.56	105783
weighted avg	0.54	0.53	0.53	105783

6. Confusion Matrix:



Model Performance

Implementing class weights was an effective tool to reduce the heavy recommendation of a single pitch type. We can see a much more even distribution of values compared to V1 in the Strategic accuracy, with better breaking and offspeed recalls. Fastball recal is lower but that is expected given we aren't recommending nearly as many fastballs. These numbers could still improve as more features are added to limit how many pitches the model has to consider (taking into consideration a pitchers arsenal) and also adding more features to track previous pitch types.

The detailed classification report also highlights the increased balance for pitche types. The main outlier is Curveballs (CU) with only 29% recall. That value is a little low, but the model may be discouraging throwing a curveball since we haven't implemented a way for it to learn its strategic benefits. For example a curveball to a hitter with a high chase rate out of the zone could be benefitial, but thats not something the model would learn based on the features we have used so far. This will be attempted in V3.

The last thing to note about the weights model is the accuracy. We do see a slight decline in Top 1 accuracy from 43% to 42%, but that is anticpated as the model is not getting lucky by guessing fastballs each time and getting it right. Seeing a slightly lower accuracy at the trade off of more evenly distriuted recommedations is reasonable.

The results for early stopping are very similar to the results from V1, which is not surprising. The purpose of early stopping here is to prevent the model from overfitting, not necessarily perform better. We still see the heavy recommendation of fastballs which is to be expected. What was valuable about early stopping is it gives us confidence that we ran enough epochs and that the model decreased the loss as much as it could.

The addition of the dropout layer is something that will be monitored as more model versions are developed. Currently, we only have a single linear layer in the RNN, which will not cause dropout to be very effective. Typically you would want several layers to drop values from at random in order to improve generalization.

The results though were not discouraging though, as we say large bumps to the Strategic Accuracy table. Breaking and Offspeed pitches went up by a good bit just from adding these layers, and fastball only dropped by about 10%. The detailed classification report was still a little sporatic, but we could see improvment to those values are more appropriate layers are added. That will be a research task in the future, to determine how many layers and what types of layers could get added to the RNN.

Finally the embedding model had improved performance as well with better strategic accuracy. I'd like to continue to play around with the embedding dimension sizes to see if there is much more impact to different values, but my approach was to scale the dimension size with the size of the feature. So batter and pitcher features have size 32 give then are a few thousand total players, where the feature for the next pitch type only has 13 options.

Future Improvements

Model V3 development will be focused around features, and figuring out what additional features could be impactful to the model.

1. Add pitch related features for previous N pitch types (start with 2, maybe 3 but no more than 3 given the size of most sequences)

- These can include what the previous N pitch types were, what the previous pitch family were, what type of count the pitch was thrown in (ahead, behind, even)

2. Also for pitchers I'd like to implement a feature for pitcher arsenals. I'm thinking of representing it in some type of vector with what pitches they can throw encoded into that vector, but i'm not sure how to go about doing and applying that to the model's learning

3. For batter's and pitchers, I want to implement features focused on their performance in certain game states. To keep things simple, I'll start by sticking to whether they were ahead, even, or behind in the count. I'm hoping this reveals pitchers are more likely to throw fastballs or pitches that are a higher strike percentage when they are behind, and breaking/offspeed pitches when they are ahead or with 2 strikes.
4. I'd also like to create some function to generate features for batter swing rates and pither usage rates. Specfically, how often does a batter swing at a certain pitch family in certain counts. Same for pitchers, how often do they throw certain types of pitches in certain counts. The goal here would be to represent the batter and pitchers specific tendencies more.

In [ ]:
```
# out_dir = Path.cwd() / "trained-parameters"
# out_dir.mkdir(parents=True, exist_ok=True)
# filename = f"simple_rnn_v0_1_{datetime.now().strftime('%Y%m%d')}.pt"
# save_path = out_dir / filename
# torch.save(model.state_dict(), save_path)
# print(f"Wrote model parameters to {save_path}")
```